

MATH 164 Discussion (Week 1): Prereqs for Optimization

Same worksheet for both sections.

Focus: limit points/closed sets, continuity, derivatives, vector calculus and why they matter for optimization.

A. Limit points, closed sets, compact sets

Limit point (accumulation point). A point x is a limit point of $S \subset \mathbb{R}^n$ if every open ball $B(x, r)$ contains a point of S different from x .

Closed set. S is closed if it contains all its limit points (equivalently: $x_k \in S$ and $x_k \rightarrow x$ implies $x \in S$).

Compact (in $\mathbb{R}^n$). S is compact iff it is **closed and bounded** (Heine–Borel).

In Optimization. Compact feasible set + continuous objective $\Rightarrow$ minimizer exists.

Quick checks: sets and limit points



1. Which sets are closed in $\mathbb{R}$? (circle: yes/no)

$(0, 1) \subset \mathbb{R}$ (closed? yes/no), $[0, 1]$ (yes/no), $\{1/n : n \in \mathbb{N}\}$ (yes/no).

2. For $S = (0, 1)$: list two limit points of S in $\mathbb{R}$. _____

3. True/False: If S is closed, then it is bounded. _____

Optimization example: existence can fail

Give an example of a continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ and a *non-compact* feasible set S such that

$$\inf_{x \in S} f(x) \text{ exists but no minimizer exists.}$$

Hint: open interval or unbounded set.

Your example: $S =$ _____ $f(x) =$ _____ Why no minimizer? _____

B. Continuity and why we care

Continuity at x . f is continuous at x if $x_k \rightarrow x \Rightarrow f(x_k) \rightarrow f(x)$.

Extreme Value Theorem (EVT). If $S \subset \mathbb{R}^n$ is compact and $f : S \rightarrow \mathbb{R}$ is continuous, then f attains both a minimum and a maximum on S .

Optimization link. Existence of solutions to constrained problems often reduces to verifying:

(i) feasibility set compact + (ii) objective continuous.

EVT for Existence of Extremizer



For each problem, state whether a minimizer is **guaranteed** by EVT, and briefly why.

1. $\min_{x \in [0, 1]} (x - 0.3)^2$ Guaranteed? _____ Reason: _____

2. $\min_{x \in (0, 1)} (x - 0.3)^2$ Guaranteed? _____ Reason: _____

3. $\min_{x \in \mathbb{R}} e^{-x^2}$ Guaranteed? _____ Reason: _____

B*. Linear algebra basics (self-check; rank/solvability/symmetry/projections/PD/norms)

- (T/F) Multiplying a column of A by a nonzero scalar does not change $\text{rank}(A)$. _____
- (T/F) Interchanging two columns could change $\text{rank}(A)$. _____
- (T/F) Adding to a column a linear combination of other columns does not change $\text{rank}(A)$. _____
- (T/F) $Ax = b$ has a solution iff $\text{rank}(A) = \text{rank}([A \mid b])$. _____
- If $A \in \mathbb{R}^{m \times n}$ and $\text{rank}(A) = m$, then the number of free variables in solving $Ax = b$ (when solvable) is _____.
- (T/F) All eigenvalues of a real symmetric matrix are real. _____
- (T/F) Any real symmetric $n \times n$ matrix has an orthogonal eigenbasis. _____
- Fill in: $\mathcal{R}(A)^\perp = \underline{\hspace{2cm}}$, $\mathcal{N}(A)^\perp = \underline{\hspace{2cm}}$.
- (T/F) P is an orthogonal projector iff $P^2 = P = P^\top$. _____
- (T/F) For symmetric Q , $Q \succ 0$ iff all eigenvalues of Q are positive. _____
- Frobenius norm:
$$\|A\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2 \right)^{1/2}.$$
 Compute $\left\| \begin{bmatrix} 1 & -2 \\ 2 & 0 \end{bmatrix} \right\|_F = \underline{\hspace{2cm}}$

Short application (optional, 2 min). Decide whether each is an orthogonal projector (yes/no) and give one reason ($P^2 = P$? symmetric?):

$$P_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \underline{\hspace{2cm}} \quad P_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \underline{\hspace{2cm}}$$

C. Derivatives, gradients, and first-order conditions

1D derivative. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

Gradient (multivariable). For $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $\nabla f(x)$ satisfies

$$f(x+d) \approx f(x) + \nabla f(x)^\top d \quad (\text{small } d).$$

First-order necessary condition (unconstrained). If f is differentiable and x^* is a local minimizer in the interior of the domain, then

$$\nabla f(x^*) = 0.$$

(Second-order: $\nabla^2 f(x^*) \succeq 0$ is typical for local minima.)

Optimization link: 1D search shows up via line search. Given direction d , define $\phi(\alpha) = f(x + \alpha d)$ and minimize over $\alpha > 0$.

Compute + interpret



Let $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with $Q = Q^\top$.

- $\nabla f(x) = \underline{\hspace{2cm}}$ $\nabla^2 f(x) = \underline{\hspace{2cm}}$
- If Q is positive definite, then f has a **unique** minimizer. Why?
- What if
 - Q is Positive Semi-Definite while not Positive Definite?
 - Q have negative eigenvalues?
 - Q is not symmetric?
- Solve $\nabla f(x) = 0$: the minimizer satisfies $Qx^* = \underline{\hspace{2cm}}$

Line-search chain rule

Let $\phi(\alpha) = f(x + \alpha d)$. Derive:

$$\phi'(\alpha) = \underline{\hspace{2cm}}$$

Then: a “best step size” often solves $\phi'(\alpha) = 0$.